

Chapter

Physical Fitness

1. Which test is conducted to assess muscle strength? MAIN 2024

- (a) Cooper run.
- (b) Harvard step test.
- (c) Push up test.
- (d) Zigzag test.

2. A 100 percent perfection at hitting the target refers to the skill component _____. SQP 2021

- (a) Speed
- (b) Strength
- (c) Accuracy
- (d) Power

3. The ability to perform a task or move from one point to another in the shortest period of time is called : COMP 2023

- (a) Power
- (b) Speed
- (c) Balance
- (d) Agility

4. Timed shuttle run is useful to improve which component of Physical Fitness? SQP 2022

- (a) Speed
- (b) Agility
- (c) Power
- (d) Accuracy

5. What is the effect of age on physical fitness? SQP 2021

- (a) Fitness levels improve automatically with age
- (b) Fitness levels remain constant regardless of age
- (c) Fitness levels tend to decline with age
- (d) Age has no impact on physical fitness

6. Which of the following activities best improves muscular strength? COMP 2024

- (a) Running a marathon
- (b) Swimming for 30 minutes
- (c) Lifting heavy weights
- (d) Playing table tennis

7. What is the main benefit of cardiovascular endurance? MAIN 2023

- (a) Increased flexibility
- (b) Enhanced lung and heart efficiency
- (c) Faster muscle growth
- (d) Improved bone density

8. Which of the following is a skill-related component of fitness? COMP 2022

- (a) Body Composition
- (b) Cardiovascular Endurance
- (c) Coordination
- (d) Muscular Strength

9. Write the importance of cardiovascular fitness. COMP 2023

10. What is the importance of muscular endurance? SQP 2022

11. Assertion (A) : Cardiovascular endurance helps in improving lung and heart efficiency.

Reason (R) : Cardiovascular endurance refers to the ability of muscles to perform repeated contractions over time. **MAIN 2024**

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.

- (c) A is true, but R is false.
 (d) A is false, but R is true.

12. Assertion (A) : Strength training helps in increasing bone density.

Reason (R) : Weight-bearing exercises stimulate bone growth and reduce the risk of osteoporosis. **SQP 2021**

- (a) Both A and R are true, and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true, but R is false.
 (d) A is false, but R is true.

13. Assertion (A) : Flexibility reduces the risk of injuries during physical activities.

Reason (R) : Improved flexibility increases the range of motion at a joint. **COMP 2023**

- (a) Both A and R are true, and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true, but R is false.
 (d) A is false, but R is true.

14. Matching Factors Affecting Physical Fitness

Column A			Column B
1.	Heredity	A.	Affects fitness due to genetic traits like height and muscle structure.
2.	Nutrition	B.	Ensures energy levels and muscle health with proper diet.
3.	Training	C.	Improves endurance, strength, and overall performance.
4.	Environment	D.	Impacts fitness through pollution and hygiene levels.

MAIN 2021

- (a) 1-A, 2-B, 3-C, 4-D
- (b) 1-B, 2-C, 3-D, 4-A
- (c) 1-D, 2-A, 3-B, 4-C
- (d) 1-C, 2-D, 3-A, 4-B

15. Identify the Fitness Component



What component of physical fitness is demonstrated in the image?

- (a) Muscular Strength
- (b) Cardiovascular Endurance
- (c) Flexibility
- (d) Power

16. What is the difference between strength and power? MAIN 2024

17. What is the meaning of Physical Fitness? SQP 2021

18. How physical fitness boosts energy in an individual? COMP 2023

19. A sedentary lifestyle leads to a reduction in _____. SQP 2021

- (a) Muscle mass and cardiovascular health
- (b) Height and flexibility
- (c) Eye coordination and digestion
- (d) Blood volume and muscle memory

20. The ____ principle states that fitness levels decline when training is stopped. COMP 2023

- (a) Overload

- (b) Reversibility
- (c) Progression
- (d) Specificity

21. What is static balance and dynamic balance? Give an example of each.
SQP 2021

22. Why is regular exercise important for mental health? COMP 2023

23. Explain the importance of posture in physical fitness. SQP 2022

24. Write the importance of Physical Fitness. MAIN 2024

25. How does sleep affect physical fitness? MAIN 2024

26. Arjun plays football regularly but often feels exhausted after a few minutes of intense running. His coach suggests he improve his ability to sustain physical activity over long durations. MAIN 2024

- (i) Which component of physical fitness does Arjun need to develop?
- (ii) What type of exercises can help him improve this component?

27. Meera is training for gymnastics and needs to improve her ability to bend, stretch, and perform movements smoothly. Her coach recommends yoga and stretching exercises. SQP 2021

- (i) Which component of physical fitness is Meera focusing on?
- (ii) Why is this component important for gymnastics?

28. Rohan wants to build his upper body strength and improve his ability to lift heavy weights. His trainer asks him to perform push-ups, weight lifting and resistance training. COMP 2023

- (i) Which component of physical fitness is Rohan working on?
- (ii) How does this component benefit his overall physical fitness?

29. Tanya, a long-distance runner, is advised to consume a well-balanced diet with high energy sources to support her endurance training. SQP 2022

(i) Which component of physical fitness is most crucial for long-distance running?

(ii) Why is proper nutrition important for enhancing this component?

Solution

1. (c) Push up test.

The push-up test is used to assess muscle strength, particularly of the upper body, including the chest, shoulders, and triceps. It evaluates the ability of muscles to exert force repeatedly over time.

2. (c) Accuracy

Accuracy refers to the ability to perform a skill with precision, such as hitting a target perfectly. A 100 percent perfection in executing the intended action indicates high accuracy.

3. (b) Speed

Speed is the ability to perform a task or cover a distance in the shortest possible time, reflecting quickness of movement and reaction.

4. (b) Agility

Timed shuttle run helps improve agility, which is the ability to quickly change direction or position while maintaining control of the body.

5. (c) Fitness levels tend to decline with age

As people get older, muscle strength, bone density, flexibility, and cardiovascular efficiency naturally decrease. Regular physical activity and proper nutrition can help slow this decline and maintain overall fitness.

6. (c) Lifting heavy weights

Muscular strength is the ability of a muscle or muscle group to exert maximum force against resistance. Lifting heavy weights or performing resistance training increases muscle fiber size and neuromuscular coordination, effectively improving strength. Endurance activities like running or swimming mainly enhance stamina, not maximum force.

7. (b) Enhanced lung and heart efficiency

Cardiovascular endurance is the ability of the heart, lungs, and blood vessels to supply oxygen to muscles during prolonged physical activity. Exercises like

running, cycling, and swimming improve heart and lung efficiency, increasing stamina and reducing the risk of cardiovascular diseases.

8. (c) Coordination

Coordination is a skill-related component of fitness that enables different body parts to work together efficiently. It is crucial for sports requiring precise movements, such as hand-eye coordination in basketball or foot-eye coordination in soccer.

9. Importance of Cardiovascular Fitness:

Cardiovascular fitness is vital for the efficient functioning of the heart, lungs, and circulatory system. It ensures proper delivery of oxygen and nutrients to muscles during physical activity, enabling sustained effort over longer periods. Benefits include:

- Reducing the risk of heart disease and high blood pressure.
- Enhancing endurance and overall stamina.
- Improving overall health and reducing fatigue.
- Supporting better performance in daily activities and sports.

10. Importance of Muscular Endurance:

Muscular endurance is the ability of muscles to perform repeated contractions over a long period without getting fatigued. Its importance includes:

- Enhancing performance in activities like cycling, swimming, and long-distance running.
- Supporting prolonged physical activity by relying on slow-twitch muscle fibers.
- Bridging the gap between cardiovascular fitness and muscular strength.
- Helping maintain stamina during extended workouts or physically demanding tasks.
- Reducing the risk of muscle fatigue and injury during continuous activity.

11. (c) A is true, but R is false.

Cardiovascular endurance enhances the efficiency of the heart and lungs by ensuring a steady supply of oxygen to the muscles during prolonged activity. However, the reason given describes **muscular endurance**, which is the ability of muscles to perform repeated contractions over time. Therefore, while the assertion is correct, the reason does not correctly explain it.

12. (a) Both A and R are true, and R is the correct explanation of A.

Strength training involves weight-bearing exercises that place stress on bones. This stress stimulates bone-forming cells, increasing bone density and strength. Regular resistance or weight training reduces the risk of osteoporosis and fractures, making the reason a correct explanation of the assertion.

13. **Answer:** (a) Both A and R are true, and R is the correct explanation of A.

Flexibility improves the range of motion at joints, allowing muscles and tendons to stretch efficiently during physical activity. This increased mobility reduces muscle stiffness and the likelihood of strains or injuries, making the reason a correct explanation of the assertion.

14. (a) 1-A, 2-B, 3-C, 4-D

Heredity influences physical traits (A), Nutrition is key for muscle health (B), Training enhances fitness (C), and Environment affects overall wellbeing (D).

15. (b) Cardiovascular Endurance

Cardiovascular endurance is the ability of the heart, lungs, and circulatory system to deliver oxygen-rich blood to muscles during prolonged physical activity. Activities like jogging, swimming and cycling improve heart function and help maintain stamina.

16. Strength is the amount of force muscles can exert against resistance for a short duration, such as lifting weights or body weight exercises. Power refers to the ability to perform maximum effort in the shortest time possible, like speeding through players in football or lifting heavy weights rapidly.

17. Meaning of Physical Fitness:

Physical fitness is the **ability of an individual to perform daily activities efficiently** with enough **energy, endurance, and strength**, while also being capable of handling **leisure activities and emergency situations**. It ensures overall **health, well-being, and functional efficiency** of the body.

18. Physical fitness boosts energy by improving circulation and reducing fatigue. Regular exercise increases energy levels, making an individual feel refreshed and active.

19. (a) Muscle mass and cardiovascular health

A sedentary lifestyle reduces muscle strength and cardiovascular efficiency. Without regular physical activity, muscles weaken, and the heart and lungs work less efficiently, increasing the risk of obesity, heart disease, and loss of endurance. Regular exercise maintains muscle tone, heart health, and overall fitness.

20. (b) Reversibility

The Reversibility Principle states that fitness levels decline when regular training is stopped. Discontinuing exercise leads to muscle loss, decreased strength, and reduced endurance. Consistent physical activity is essential to maintain the gains achieved through training.

21. (i) **Static Balance** – The ability to maintain stability while staying still.

Example: Holding a yoga pose like the Tree Pose.

(ii) **Dynamic Balance** – The ability to maintain stability while in motion.

Example: Riding a bicycle or walking on a balance beam.

22. Regular exercise is important for mental health because it stimulates the release of **endorphins**, known as “feel-good hormones.” These hormones help reduce **stress, anxiety, and depression**, while improving **mood, concentration, and overall cognitive function**, leading to better mental well-being.

23. Importance of Posture in Physical Fitness:

Good posture is essential for maintaining **proper spinal alignment**, which helps the body function efficiently during physical activities. It **reduces strain on joints and muscles**, prevents muscle imbalances, and **lowers the risk of injuries**, especially back and neck pain. Maintaining correct posture also improves **balance, stability, and overall body mechanics**, enhancing performance in exercises and daily tasks.

24. Importance of Physical Fitness:

1. **Overall Health:** Enhances general health by improving blood circulation, boosting immunity, and reducing risks of lifestyle diseases like obesity, diabetes, and heart problems.
2. **Energy Boost:** Increases energy levels, reduces fatigue, and helps individuals stay active and productive throughout the day.
3. **Weight Management:** Helps maintain ideal body weight by burning extra calories and improving metabolism, preventing obesity and related complications.
4. **Strong Build:** Strengthens muscles and bones, reducing the risk of injuries, back pain, and osteoporosis.
5. **Mental Strength:** Improves blood flow to the brain, enhancing memory, focus, and mental resilience. Regular exercise also helps manage stress and maintain emotional stability.
6. **Personality Development:** Enhances appearance, confidence, and self-esteem, promoting a positive and optimistic outlook on life.

25. Sleep plays a crucial role in maintaining physical fitness by:

1. **Muscle Recovery:** Allows muscles to repair and grow after exercise, enhancing strength and endurance.
2. **Energy Restoration:** Replenishes energy stores, ensuring optimal performance during physical activities.

3. **Hormone Balance:** Regulates hormones like growth hormone and cortisol, which are essential for metabolism, muscle growth, and stress management.
4. **Performance Impact:** Lack of sleep leads to fatigue, slower reaction times, reduced coordination, and decreased overall athletic performance.

26. (i) Cardiovascular Endurance – This component of physical fitness allows an individual to sustain prolonged physical activity by improving the efficiency of the heart, lungs, and circulatory system.

(ii) Exercises to improve cardiovascular endurance:

- Long-distance running
- Swimming
- Cycling

These activities enhance oxygen delivery to muscles, increase stamina, and reduce fatigue during extended physical activity.

27. (i) Flexibility – It enhances the range of motion at joints, allowing smoother and more controlled movements.

(ii) Importance for gymnastics:

Flexibility is crucial for gymnastics because it enables movements such as bending, stretching, twisting, and balancing. Improved flexibility reduces the risk of injuries and helps gymnasts perform complex routines with precision and grace.

28. (i) Muscular Strength – The ability of muscles to exert force against resistance, developed through activities like push-ups, weight lifting, and resistance training.

(ii) Benefits:

- Enhances the capacity to perform heavy physical tasks.
- Improves posture and body alignment.

- Reduces the risk of muscle strains and related injuries.
- Supports overall physical performance and functional fitness.

29. (i) Cardiovascular Endurance – This allows Tanya to sustain prolonged running by efficiently supplying oxygen to her muscles.

(ii) Importance of Proper Nutrition:

- Provides sufficient energy to meet the demands of long-distance running.
- Maintains stamina and delays fatigue.
- Supports muscle recovery and repair after training.
- Enhances overall endurance performance and efficiency.